



DATA OPERATIONS / CSV INTELLIGENCE

Inconsistent source data, made predictable.

A practical data-quality demonstration showing how inconsistent names, prices, statuses, dates, duplicates, and missing values become a canonical dataset while source provenance and exceptions remain reviewable.

CANONICAL DATA PIPELINE

Messy rows -> governed schema.

RAW	CANONICAL
PRICE \$19.95	PRICE 19.95
STATUS in stock	STATUS AVAILABLE
DATE 8/7/26	DATE 2026-08-07
VENDOR BLUE RIDGE	VENDOR Blue Ridge

QUALITY FLAGS STAY VISIBLE

Duplicates / missing values / provenance remain visible for review.

INDEPENDENT PROJECT / DATA OPERATIONS DEMONSTRATION
PUBLIC-FACING, SANITIZED PROJECT DEMONSTRATION

- CANONICAL SCHEMA
- SOURCE PROVENANCE
- QUALITY FLAGS
- NO SILENT GUESSES



CSV Normalization & Data Quality Control

PROBLEM

Operational CSV files often mix naming conventions, data types, date formats, duplicates, missing fields, and ambiguous statuses. Silent cleanup can destroy provenance or introduce unsupported assumptions.

APPROACH

Preserve the raw row, map fields into a canonical schema, standardize types and enumerations, detect duplicates, flag missing or conflicting values, and retain the source identity required for review and downstream workflows.

Result: source values remain traceable while downstream fields become predictable.

The example shows a deliberately small synthetic CSV slice so the transformation is visible at a glance.

BEFORE - INCONSISTENT SOURCE

ID	VENDOR	PRICE	STATUS
001	BLUE RIDGE	\$19.95	in stock
001	Blue Ridge	19.95	Y
003	nova-supply	19,95	available
004	NOVA SUPPLY		unknown
005	Nova Supply	\$22	out

AFTER - CANONICAL + QA STATE

ID	VENDOR	PRICE	STATUS
001	Blue Ridge	19.95	AVAILABLE
001-DUP	Blue Ridge	19.95	DUPLICATE
003	Nova Supply	19.95	AVAILABLE
004	Nova Supply	NULL	REVIEW
005	Nova Supply	22.00	UNAVAILABLE

TRANSFORMATION RULES

Preserve source row -> standardize types -> normalize enums -> detect duplicates -> flag missing values -> retain provenance.

CAPABILITIES DEMONSTRATED

Data Cleaning / CSV Handling / Schema Design / Data Quality / Duplicate Detection / QA Documentation

